REMIND: Recall Enhanced Memory Integration for Natural Language Dialogue Systems

Bo-Sheng Huang*, Ted T. Kuo*, Li-Jen Wang*, Chia-Yu Lin†

*College of Artificial Intelligence, National Yang Ming Chiao Tung University, Hsinchu, Taiwan
Email: {carbarcha.ai12, tedkuo, edwardwang}@nycu.edu.tw
†Department of Computer Science and Information Engineering, National Central University, Taoyuan, Taiwan
Email: sallylin0121@csie.ncu.edu.tw

***Abstract*—Large Language Models (LLMs) lack robust memory management for multi-turn dialogues, limiting their effectiveness in personalized applications. We introduce REMIND, a lightweight, modular framework integrating Short-Term Memory (STM) for session-specific recall and Long-Term Memory (LTM) for persistent knowledge retention. REMIND employs a hybrid retrieval mechanism that combines vector search, hierarchical grouping, iterative query refinement, and keyword filtering to enhance recall precision without modifying LLM architectures. To benchmark REMIND, we introduce MemoRA, an open dataset for evaluating dynamic memory retrieval in multi-turn dialogues. Our evaluation combines static metrics with Kernel Density Estimation (KDE) and `GPT-4o-mini` as a relevance judge, showing that REMIND outperforms MemGPT in recall scenarios. Results indicate that `LLaMA3.3:70B` demonstrates weaker grouping capabilities, while `DeepSeek-V3` is a cost-effective reasoning engine. By open-sourcing the MemoRA dataset [1] and REMIND's implementation [2], we provide resources to advance research in memory-augmented dialogue systems, enabling personalized, scalable applications in healthcare, tutoring, and customer support. While current evaluations focus on the MemoRA and DMR datasets, future work will extend assessments to broader benchmarks.**

***Index Terms*—Memory-Augmented Dialog Systems, Large Language Models (LLMs), Hybrid Memory Retrieval, MemoRA dataset, Kernel Density Estimation (KDE), Multi-Turn Dialogue Systems, Personalized Memory Systems.**

I. Introduction

Transformer architectures have revolutionized natural language processing (NLP), enabling LLMs to excel in various domains such as healthcare, education, and legal advising. However, LLMs are limited by their inability to retain memory across interactions, relying solely on query-time inputs, which impairs their effectiveness in personalized, context-aware applications.

Existing solutions, such as expanded context windows [3] and compressive memory techniques [4], partially address memory limitations but remain computationally intensive and susceptible to phenomena like *lost-in-the-middle* [5]. Similarly, external memory systems like Retrieval-Augmented Generation (RAG) [6] emphasize general-purpose retrieval, overlooking user-specific or session-based memory needs critical for personalized applications. Similarly, systems like MemGPT [7] simulate memory through function calls, but their reliance on resource-intensive, proprietary models like GPT-4 limits scalability.

To address these challenges, we introduce REMIND, a modular framework enabling LLMs to integrate STM for session-specific interactions and LTM for persistent knowledge retention. Using vector-based similarity search, hierarchical grouping, and iterative query refinement, REMIND ensures precise and contextually coherent memory retrieval without modifying LLM architectures.

REMIND achieves scalability and adaptability through prompt engineering techniques tailored for session-specific and long-term memory retrieval. KDE visualizations (Figs. 2 and 3) highlight the framework's ability to evaluate retrieval quality dynamically, bridging the gap between session-specific and historical context retrieval.

Contributions:

- **REMIND Framework:** A lightweight, modular system enabling LLMs to integrate STM and LTM for dynamic memory management without altering their architecture.
- **MemoRA Dataset:** An open benchmark dataset [1] for evaluating memory retrieval in multi-turn dialogues, enabling comprehensive assessments of memory-augmented systems.
- **Evaluation Techniques:** A novel methodology combining static metrics with KDE [8] and an LLM-based judge to evaluate short- and long-term memory retrieval strategies.
- **LLM Evaluation Insights:** Experiments reveal significant variability in grouping quality among LLMs, with `LLaMA3.3:70B` [9] underperforming compared to `DeepSeek-V3` [10] and `GPT-4o-mini`, emphasizing the need for consistent judging criteria and model-specific evaluation protocols.
- **Cost-Effective Reasoning Engine:** Identification of `DeepSeek-V3` as a robust and cost-effective reasoning engine that enhances REMIND's memory retrieval capabilities.

II. Related Work

Large language models (LLMs) built on transformer architectures excel in natural language understanding, reasoning, and task-solving. However, their reliance on static pre-trained

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knowledge and fixed context windows limits their effectiveness in tasks requiring long-term memory, personalization, or contextual continuity. Prior work in retrieval-augmented frameworks, memory-augmented systems, and hybrid retrieval mechanisms provides valuable insights but reveals critical limitations.

A. Retrieval Augmented Generation (RAG) Frameworks

Retrieval Augmented Generation (RAG) frameworks [6], [11] enable LLMs to dynamically retrieve external knowledge during inference without altering pre-trained parameters, proving effective in tasks like open-domain question answering. However, they primarily rely on static datasets like Wikipedia and lack mechanisms for handling user-specific or session-based memory, which is crucial for personalized applications such as tutoring or healthcare. RAG frameworks also struggle with maintaining coherence in multi-turn dialogues, often retrieving isolated information without considering sequential dependencies [12]. To address these limitations, REMIND focuses on session-adaptive, personalized retrieval without modifying LLM architectures, distinguishing itself from RAG and MemGPT.

B. Architectural Modifications for Memory-Augmented Systems

Memory-augmented systems improve memory capabilities through architectural changes, such as memory layers [13] and compressive transformers [4]. These methods enhance efficiency and scalability by introducing trainable memory mechanisms or extending context windows with compressive memory modules [3]. However, they require substantial retraining and are less adaptable to emerging models, making them resource-intensive and impractical for real-world applications demanding modularity.

C. Hybrid Retrieval Mechanisms and Memory Organization

Hybrid retrieval mechanisms combine strategies like vector-based similarity search [14], re-ranking [15], and adding hypothetical document embeddings (HyDE) [16] or rewriting the user's query to refine retrieval accuracy. Introducing hierarchical memory structures further improves efficiency by organizing memory into summaries and raw records [12], [17]. However, these approaches often overlook the sequential nature of multi-turn dialogues and are typically designed for static datasets, making them unsuitable for adaptive, user-specific applications. Computational inefficiencies also limit scalability for broader deployment.

D. Need for a New Approach

Existing methods fail to address three critical challenges:

1) Retaining and retrieving user-specific or session-based knowledge.
2) Ensuring contextual coherence in multi-turn dialogues.
3) Achieving scalability and adaptability without extensive retraining.

We present a modular framework, *REMIND* (Recall-Enhanced Memory Integration for Natural Language Dialogue Systems), that dynamically integrates short-term and long-term memory to address these challenges, providing scalability, personalization, and efficient context-aware retrieval for dynamic, user-centric applications.

III. Methodology

The REMIND framework enhances LLMs by integrating hierarchical memory systems and a Language Model (LM)-driven hybrid retrieval mechanism, as shown in Fig. 1. Without modifying the LLM(processor) architecture, REMIND leverages external memory storage and prompt engineering to simulate dynamic memory, enabling user-specific context retrieval beyond native limitations.

REMIND consists of three components: *Short-Term Memory (STM)*, *Long-Term Memory (LTM)*, and a hybrid retrieval mechanism. STM stores session-specific dialogs that are later transferred to LTM for persistent, hierarchical storage. LTM organizes user-specific knowledge with timestamps, maintaining interaction sequences to enhance retrieval quality. The hybrid retrieval mechanism combines vector-based similarity with keyword frequency filtering to further refine retrieval precision, enhancing both semantic and literal relevance.

This design enables efficient, adaptive memory management, ensuring personalized, context-aware applications with high accuracy, relevance, and temporal coherence.

A. System Architecture and Workflow

The REMIND framework integrates STM, LTM, and LM-driven hybrid retrieval mechanisms to enhance LLMs' memory capabilities for multi-turn, user-specific interactions. The workflow, illustrated in Fig. 1, operates as follows:

1) A user query triggers REMIND to retrieve user-specific context.
2) STM and LTM process the query concurrently to retrieve relevant prior interactions.
3) Retrieved information from STM and LTM is combined with the query into a unified prompt and sent to the LLM(processor)'s context window (*scratchpad*) for response generation.
4) When the LLM(processor)'s context window nears its limit, the content is summarized and prepended to the LLM(processor)'s context window by the STM, while the detailed interactions remain in STM's store.
5) At session end, STM records are transferred to LTM, where they are hierarchically grouped and summarized by the LM for long-term storage.

B. Short-Term Memory (STM)

STM manages session-specific conversational data as a temporary buffer for recent interactions. It provides fast, accurate retrieval and ensures coherence during ongoing sessions.

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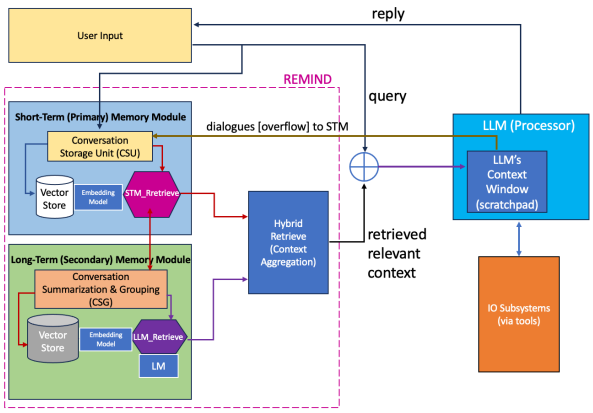

Fig. 1: REMIND System Architecture integrating STM, LTM, and LM-driven hybrid retrieval mechanisms for hierarchical user-specific memory management.

Algorithm 1 Short-Term Memory Retrieve (STM_Retrieve)

procedure STM_RETRIEVE($\mathbf{q}$, k)
  $\mathbf{q}_{\text{vec}} \leftarrow$ EMBEDQUERY($\mathbf{q}$)
  $retrieved \leftarrow$ SIMSEARCH($\mathbf{q}_{\text{vec}}, \mathcal{C}_{\text{STM}}, k$)
  $sorted \leftarrow$ SORTBYTIMESTAMP($retrieved$)
  return $sorted$
end procedure

1) STM Memory Management: STM maintains a sliding window of recent conversations:

$$\mathcal{C}_{\text{STM}}(t) = \{c_{t-w+1}, \ldots, c_t\},$$

where c_t is the user-assistant conversation at time t and w is the window size. When the number of stored interactions exceeds w, STM generates a summary of the prior interactions as a compressed history:

$$\mathbf{s}_{\text{STM}} = g_{\text{summary}}(\mathcal{C}_{\text{STM}}),$$

and appends it to the LLM's(processor) context window while the detailed records are vectorized using an embedding model and preserved in the vector store of STM for efficient retrieval.

2) STM Retrieval: STM employs vector-based similarity search to retrieve the top-k relevant interactions for a query $\mathbf{q_{vec}}$:

$$\text{Rel}_{\text{STM}} = \text{top}_k(\text{sim}(\mathbf{q_{vec}}, \mathcal{C}_{\text{STM}})), \tag{1}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity. To ensure chronological coherence, retrieved results are sorted by timestamp, as shown in Alogrithm 1.

C. Long-Term Memory (LTM)

LTM provides persistent storage for user-specific knowledge by hierarchically organizing conversations into topic-based groups. Each group contains conversations related to the same topic and a summary generated by the LM to capture their semantic essence.

When transferring records from STM to LTM, conversations are classified and assigned to the most relevant group. A new one is created with a corresponding summary if no suitable group exists. Group summaries are dynamically updated to reflect new additions, ensuring scalability and efficient retrieval of detailed context and high-level summaries.

This structure enables LTM to maintain continuity across sessions, supporting personalized and context-aware responses while ensuring efficient retrieval and storage.

1) LTM Memory Management: When a session ends, conversations from STM are classified by the LM and added to the most relevant group. If no suitable group exists, a new one is created with a summary capturing its semantic essence. Groups dynamically update their summaries as new conversations are added, maintaining scalability and retrieval efficiency.

2) LTM Retrieval: LTM retrieval is a multi-step process that balances contextual coherence and efficiency:

a) *Select Top-k Groups by Relevance*: Group summaries, $\mathbf{g}_{sumry}$, are ranked by their similarity to the query vector $\mathbf{q_{vec}}$ using cosine similarity:

$$\text{Rel}_{\text{grps}} = \text{top}_k(\text{sim}(\mathbf{q_{vec}}, \{\mathbf{g}_{sumry,j}\})) \;\forall\; j \in k\text{-grps} \tag{2}$$

b) *Retrieve Conversations from the Top Group*: Within the most relevant group G, top-r conversations are retrieved:

$$\text{Rel}_{LTM_{\text{G, child}}} = \text{top}_r(\text{sim}(\mathbf{q_{vec}}, \mathcal{C}_G)). \tag{3}$$

c) *Jump to Another Group*: If the current group is insufficiently relevant, the LM could explore another group from the top-k list based on their group summary and start the next retrieve round.

d) *Rewrite the Query*: If there is no adequate context, the LM can also decide to rewrite the query based on search history to initiate a new retrieval process.

e) *Combine Results*: In each round, the retrieved context constitutes group summaries Eq. (2), the top-group's top-r conversations Eq. (3), and search history:

$$\mathcal{R}_{\text{LTM}} = \text{sort}(\text{Rel}_{\text{grps}} \cup \text{Rel}_{LTM_{G,\text{child}}}). \tag{4}$$

Results are chronologically ordered to preserve interpretability and coherence. The retrieval process ends if the LM finds enough context or the max rounds have been reached.

3) Hybrid Retrieval Mechanisms: STM and LTM are queried concurrently to retrieve the most relevant context (Eqs. (3) and (4)) for a user query. Retrieved results are merged into a unified prompt:

$$\mathcal{P} = g_{\text{combine}}(\text{Rel}_{\text{STM}}, \mathcal{R}_{\text{LTM}}, \mathbf{q}),$$

which is then passed to the LLM (processor) for downstream tasks:

$$resp_t = f_{\text{LLM}}(\mathcal{P}).$$

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4) Balancing Exploration and Exploitation: Effective memory retrieval requires balancing exploitation (leveraging recent, contextually relevant information) and exploration (considering deeper historical interactions). For example, when asked, "Japanese or Chinese for lunch?" a recent conversation might state, "I want sushi, not a burger, for lunch," suggesting a preference for Japanese cuisine. However, a record from three weeks prior stating, "When it comes to food, I am all for Chinese," reveals a broader preference. A retriever that over-relies on the most recent context risks missing critical, long-term preferences. Our approach ensures both recent and older contexts are evaluated to provide coherent and well-informed responses.

The REMIND framework, as outlined in Algorithm 2, addresses key limitations of LLMs' context window sizes by enabling scalable and efficient storage and retrieval through hierarchical memory organization. It maintains contextual coherence via timestamp-based sorting and ensures adaptability with a modular design. REMIND's hybrid retrieval mechanism facilitates dynamic and precise access to short-term and long-term memories, enabling personalized, contextually coherent, and scalable responses for diverse user-centric applications.

Algorithm 2 LTM Retrieval Algorithm

procedure LTM_RETRIEVE(q, k, r, $maxRounds$)
  $rounds \leftarrow 0$ ▷ Count retri rounds
  $visitedGrps \leftarrow \emptyset$ ▷ Track visited grps
  $sHist \leftarrow []$ ▷ Store search history
  $resultInfo \leftarrow \emptyset$ ▷ Final result to return
  while $rounds < maxRounds$ **do**
    $topKGrps \leftarrow \text{sim}_k(q, \text{all grp summaries}) \setminus visitedGrps$
    if $topKGrps = \emptyset$ **then**
      break ▷ No more grps to process
    end if
    for $grp \in topKGrps$ **do**
      $rConvs \leftarrow \text{sim}_r(q, \text{grp.convs})$ ▷ Top-r convs from grp
      $otherSummaries \leftarrow$ summaries of $topKGrps \setminus \{grp\}$
      $retrievedInfo \leftarrow \{\text{grp.summary}, rConvs, otherSummaries, sHist\}$
      $action, newHist \leftarrow \text{LM_Action}(q, retrievedInfo, sHist)$
      $sHist.\text{add}(newHist)$ ▷ Update search history
      if $action =$ "END" **then**
        $resultInfo \leftarrow retrievedInfo$
        break ▷ Sufficient info found, stop search
      else if $action =$ "JUMP" **then**
        $visitedGrps \leftarrow visitedGrps \cup \{grp\}$
        continue ▷ Move to nxt grp in top-k
      else if $action =$ "REWRITE" **then**
        $q \leftarrow \text{newHist.query}$ ▷ Update query for nxt round
        $visitedGrps \leftarrow \emptyset$ ▷ Reset visited grps for new top-k
        break ▷ Restart with new query
      end if
    end for
    if all grps in $topKGrps$ processed **then**
      break ▷ Stop if all top-k grs have been visited
    end if
    $rounds \leftarrow rounds + 1$ ▷ Incr rnd cntr
  end while
  return $resultInfo$ ▷ Return retrieved info or empty result
end procedure

IV. EXPERIMENTS

We evaluate the *LTM_Retrieve* component of the REMIND framework to assess its retrieval accuracy, contextual coherence, and adaptability for memory-intensive tasks. Four LLMs are selected as the reasoning engine, i.e., the **LM** in Fig. 1, facilitating conversation grouping and retrieval processes. `GPT-4o-mini` evaluates each LLM's grouping and retrieval capabilities as a consistent judge. Retrieval performance is assessed using relevance and coherence scores (1–10) assigned by `GPT-4o-mini`. Semantic embeddings of 1536 dimensions are generated using OpenAI's `text-embedding-3-small` model.

We compare *LTM_Retrieve* with *Base*, *Base_Paragraph*, *STM*, and *LTM_Group*. The *Base* method is similar to *STM*, as described in Algorithm 1, but excludes timestamping for conversations. *Base_Paragraph* extends *Base* by storing only summaries generated from every five conversations rather than individual interactions. The key distinction between *LTM_Retrieve* and *LTM_Group* lies in their retrieval approaches: *LTM_Retrieve* uses an iterative, multi-round process, incorporating actions like 'JUMP' to switch groups or 'REWRITE' to refine queries for enhanced relevance. In contrast, *LTM_Group* performs single-round retrieval, focusing on retrieving group summaries and the top-*r* conversations without iterative refinement.

A. Datasets and Methods

MemoRA Dataset. Unlike the DMR dataset used by MemGPT, MemoRA benchmarks both short-term and long-term memory retrieval in multi-turn dialogues, focusing on dynamic coherence across historical and session-specific contexts. Derived from the *Conversation Chronicles* [18], this dataset comprises 500 QA pairs designed to evaluate multi-turn dialogue coherence, context-dependent queries, and memory-intensive retrieval tasks. It tests the system's ability to leverage both historical and session-specific context.

DMR Dataset. Adapted from the *Multi-Session Chat* corpus, the DMR dataset [7] focuses on session-specific recall. Its 500 QA pairs require precise memory retrieval of session-specific details, making it suitable for evaluating targeted retrieval performance.

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B. Retrieval Evaluation.

Retrieval quality is assessed across five retrieving mechanisms: *Base_Dialog*, *Base_Paragraph*, *STM*, *LTM_Group*, and *LTM_Retrieve*. Scores for relevance and consistency (ranging from 1–10) are evaluated using `GPT-4o-mini`, and Kernel Density Estimation (KDE) visualizes score distributions.

Table I presents retrieval performance across different REMIND configurations, providing a controlled comparison of system variations. *LTM_Retrieve* achieved the highest relevance scores (mean: 8.42, median: 9.0) graded by `GPT-4o-mini`, demonstrating superior contextual coherence. KDE visualizations in Fig. 2 reveal a sharp peak near 9.0 for *LTM_Retrieve*, indicating its high consistency and relevance across multiple retrieval rounds. This supports REMIND's ability to maintain contextual coherence in dynamic, memory-intensive scenarios.

TABLE I: Retrieval Performance on The MemoRA Dataset (`GPT-4o-mini`).

Retriever	Mean	Median	Std. Dev.	Variance
Base_Paragraph	6.65	7.00	1.07	1.14
Base_Dialog	7.75	8.00	1.05	1.11
STM	7.79	8.00	1.10	1.20
LTM_Group	8.14	8.00	1.29	1.65
LTM_Retrieve	**8.42**	**9.00**	1.45	2.10

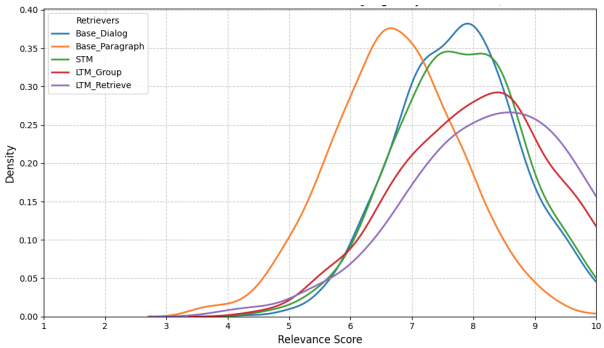

Fig. 2: Retrieve's KDE Distributions Scored by GPT-4o-mini.

Grouping Performance Across LLMs. To evaluate grouping quality, `GPT-4o-mini` was used as a consistent relevance judge. Grouping was conducted by different LLMs, including `GPT-4o-mini`, `DeepSeek-V3`, `LLaMA3.3:70B`, and `Qwen2.5:32B`, acting as the LM of the REMIND framework. Table II summarizes the results, where `DeepSeek-V3` and `GPT-4o-mini` demonstrated higher relevance scores for both *LTM_Retrieve* and *LTM_Group*. By contrast, `LLaMA3.3:70B` scored lower, highlighting limitations in structuring multi-turn conversations effectively. While model comparisons involve varying scales (e.g., LLaMA3.3:70B vs. DeepSeek-V3), our focus is on practical retrieval performance rather than direct architectural equivalence. Fig. 3 shows KDE distributions, emphasizing *LTM_Retrieve*'s broader variability and stronger peak relevance, further validating its iterative retrieval approach.

TABLE II: Performance of Different LLMs as REMIND's Grouping LM (MemoRA Dataset).

LM	Retriever	Mean	Median	Std. Dev.
GPT-4o-mini	LTM_Group	8.14	8.00	1.29
	LTM_Retrieve	**8.42**	**9.00**	1.45
DeepSeek-V3	LTM_Group	8.24	8.00	0.87
	LTM_Retrieve	**8.51**	**9.00**	1.25
Qwen2.5:32B	LTM_Group	8.13	8.00	1.16
	LTM_Retrieve	**8.23**	8.00	1.16
LLaMA3.3:70B	LTM_Group	7.38	8.00	1.34
	LTM_Retrieve	6.48	7.00	2.41

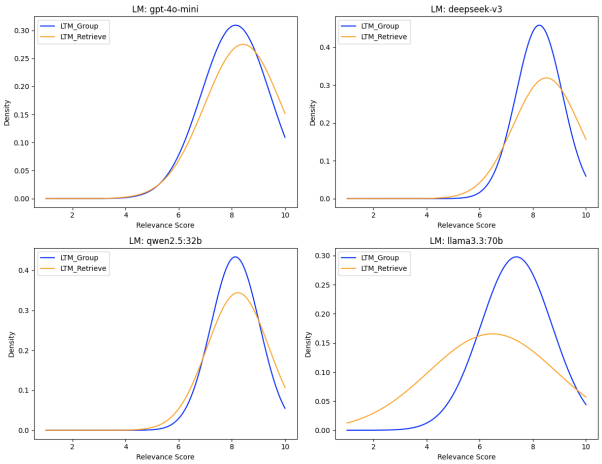

Fig. 3: KDE Distributions: LTM_Group vs. LTM_Retrieve.

Robustness of Grouping and Structuring Tasks. Using the MemoRA dataset, `GPT-4o-mini` and `DeepSeek-V3` act as the LM to group and organize retrieved conversations into JSON structures. Both performed well, with minimal retries. `DeepSeek-V3` showed slightly better robustness (1.01 retries vs. 1.02 for `GPT-4o-mini`), reflecting its cost-effectiveness.

TABLE III: Retry Rates for Grouping and Structuring Tasks.

LM	Avg. Retries
GPT-4o-mini	1.02
DeepSeek-V3	**1.01**

Retrieval on DMR Dataset. Comparisons with MemGPT using the session-specific DMR dataset reveal *STM* and MemGPT achieved the highest ROUGE-L recall scores (0.83), demonstrating strong session-specific retrieval. While *LTM_Retrieve* scored slightly lower (0.76) on session-specific tasks, REMIND's hybrid strategy effectively balances session-specific and long-term retrieval, demonstrating adaptability across diverse memory retrieval scenarios. By integrating *STM* for immediate context and *LTM_Retrieve* for persistent

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knowledge retention, REMIND overcomes the limitations of relying solely on either short- or long-term memory systems, offering a versatile solution for memory-intensive applications.

TABLE IV: Retrieval Performance on DMR Dataset (ROUGE-L Recall).

Method	ROUGE-L(R)
STM	**0.83**
LTM_Group	0.80
LTM Retrieve	0.76
MemGPT	**0.83**

C. Summary and Discussion

The experiments validate *LTM_Retrieve* as a robust and effective component of the REMIND framework, excelling in retrieving long-term context while maintaining cross-session coherence. Key findings include:

- **Grouping Effectiveness:** *LTM_Retrieve* consistently achieved higher relevance scores compared to *LTM_Group*, with `GPT-4o-mini`, `DeepSeek-V3`, and `Qwen2.5:32B` demonstrating superior grouping capabilities, while `LLaMA3.3:70B` underperformed, highlighting its limitations in structuring multi-turn conversations. KDE visualizations in Figs. 2 and 3 further underline *LTM_Retrieve*'s broader relevance consistency compared to *LTM_Group*, emphasizing its suitability for dynamic memory-intensive tasks.
- **LLM Variability:** Differences in grouping performance across LLMs underscore the challenges of relying on diverse models for memory organization, emphasizing the importance of understanding and leveraging model-specific strengths.
- **Hybrid Strategy:** The combination of *STM* and *LTM_Retrieve* balances short-term and long-term memory retrieval effectively, adapting well to the requirements of diverse datasets such as REMIND and DMR.

These findings validate the adaptability and scalability of the REMIND framework for memory-intensive tasks and highlight the importance of selecting LLMs with strong grouping and retrieval capabilities to maximize performance.

V. Conclusion

We present REMIND, a lightweight, modular framework that enables LLMs to manage dynamic memory through STM for session recall and LTM for persistent knowledge retention. Evaluated on the MemoRA dataset, REMIND demonstrates strong multi-turn and cross-session retrieval while maintaining cost-effectiveness and adaptability. By integrating a hybrid retrieval mechanism and KDE-based evaluation, REMIND provides deeper insights into retrieval quality, positioning itself as an application-level memory infrastructure that complements foundational AI systems.

Experiments using `GPT-4o-mini` reveal significant variability in LLM grouping performance. `DeepSeek-V3` exhibits strong grouping capabilities, whereas `LLaMA3.3:70B` shows greater variability and limitations. KDE visualizations highlight the importance of grouping quality in retrieval relevance, emphasizing the need for informed LLM selection in memory-intensive applications.

These findings demonstrate the need for robust evaluation frameworks and model selection strategies to improve retrieval coherence and performance. While REMIND demonstrates strong retrieval capabilities, its scalability across broader tasks remains future work.

By open-sourcing MemoRA and REMIND's implementation, we aim to foster research in memory-augmented dialogue systems, enabling scalable applications in healthcare, tutoring, and customer support.

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